Physics of Fluids

RESEARCH ARTICLE | JUNE 21 2023

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Physics of Fluids 35, 067124 (2023)
https://doi.org/10.1063/5.0151455

Physics of Fluids ARTICLE pubs.aip.org/aip/pof

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Cite as: Phys. Fluids 35, 067124 (2023); doi: 10.1063/5.0151455
Submitted: 22 March 2023 · Accepted: 2 June 2023 ·
Published Online: 21 June 2023

View Online Export Citation CrossMark

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ABSTRACT

Recently, the development of environmentally friendly refrigerants has emerged as a major issue in industry. Owing to the high cost of developing new refrigerants, falling-film evaporators are attracting increasing interest, given their reduced refrigerant usage compared with flooded evaporators. In falling–film evaporators, the distributor at the top of the evaporator uniformly delivers refrigerant through multiple holes over heat transfer tubes. We focus on the liquid (refrigerant) behavior in the distributor of a falling-film evaporator. To understand the underlying mechanism, we perform experiments on single hole as well as multiple holes in the distributor. For both cases, we have derived theories that predict the refrigerant height in the distributor, according to the flow rate. They show good agreement with the experimental measurements. By changing the hole diameter, number of holes, and flow rate, we quantitatively characterize parametric dependencies that determine the liquid behavior. In addition, we unveil a negative effect of the supply stream from the upper distributor that locally mitigates the outflow rate from the distributor.

I. INTRODUCTION

Over the years, with improving quality of life, the demand for refrigeration systems has rapidly increased. Accordingly, the demand for industrial chillers that are responsible for cooling buildings is also increasing. However, owing to the environmental problems caused by conventional refrigerants, air conditioning industries are focused toward developing refrigerants with a low global warming potential (GWP).[1] Furthermore, as the 2016 Kigali amendment was proposed to reduce the production and consumption of hydrofuorocarbons (HFCs), the development of coolers using low GWP refrigerants has become essential.[2] Given the high cost of industrial coolers that use low GWP refrigerants, the minimization of refrigerant usage is becoming increasingly attractive.

Shell and tube evaporator is one of the most widely used evaporators.[3–6] Although flooded evaporators have been commonly used, the interest in falling-film evaporators has gradually increased, owing to the reduced refrigerant requirement. In a falling-film evaporator, a distributor, called distribution tray, is an essential component that uniformly delivers refrigerant. Typically, the distributor is located at the top of the heat transfer tube bundle. If the distributor is large, the space available for the bundle decreases. Therefore, a reasonably sized distributor is required, and the distributor design in falling-film evaporators has received important research attention. The size of the distributor is typically determined by the length and width of the tube group and internal refrigerant level. Hence, accurately determining and setting the level of refrigerant are essential, because if the level is very low, the distributor is exposed to water level fluctuation problems caused by external factors. Therefore, the factors that affect refrigerant levels in distributor should be clearly understood to achieve high performance and reliability.

Despite the importance, the studies on the distributors have not been intensively performed yet. Most of the distribution trays are introduced secondarily in the some studies about the heat transfer performance.[11–14] Regarding fundamental theories, some studies associated with the classic Torricelli's theory were performed several times.[7,9,10] However, we found the liquid behaviors of low flow rate are not completely followed by the existing theory. The mechanism has not been sufficiently revealed yet. Considering actual application in falling-film evaporator (shell and tube type), the characteristics of the falling liquid at low flow rate is very important. Indeed, in various actual situations, the evaporator does not operate only in the ideal state where the refrigerant is sufficiently supplied to all parts of the distributor. For example, when a distributor is not installed perfectly horizontally, the flow rate on one side can be considerably lower than that on the opposite side because of the inclination.[15] The inclination causes a significant problem, because the large length of the evaporator results in large surface level drops even at small inclinations.

In this paper, we introduce liquid behaviors in distributors under increasing flow rate, considering measured liquid surface levels and outflow characteristics. Employing measurements for a distributor with a single hole, we obtain the liquid behavior on a distribution tray with multiple holes. In addition, we aim to introduce explanatory theories, including an analytical solution for the equilibrium surface level to further understand and describe the experimental measurements.

II. EXPERIMENTAL SETUP

Two different distributors are widely used in industry, as shown in Fig. 1, and both depend on gravitational force to operate. Their structural properties, such as hole size and pitch, can be considered as control variables. In the figure, the first type is employed in absorption chillers, which use water as the refrigerant and an aqueous solution of lithium bromide (LiBr) as the absorbent[11,12] [see Fig. 1(a)]. The major advantage of this type is that it can supply liquid to the exact location even at a low flow rate. The sharp triangular guides assist in supplying liquid at a specific point against capillary force that deflects the liquid stream at a low flow rate. Thus, this type is used for liquids with high surface tension. We note that the surface tension of the LiBr solution is even higher than water.

The second type has numerous circular holes at the bottom plate and is widely used in falling-film evaporators owing to its simplicity[16] [see Fig. 1(b)]. In addition to the simplicity of the structure, another advantage of the second type is that it takes up less space than the first type. The space occupied by the distributor is directly related to the total size of the evaporator, which is very important for product sales.

In this study, we address the general refrigerant behavior characteristics of distributors. Since flow in the first type is always associated with its own complex structure, we focus on the second type for better understanding in this study.

In experiments, visualization analysis is performed to reveal the flow mechanism in the distributor. Although an actual distributor has a number of holes on the bottom plates, we perform experiments on single holes as well as multiple holes, to focus on the liquid behaviors more accurately. In fact, the flow through a hole is affected by that through adjacent holes. For example, the flow rate and direction may be changed depending on the flow through adjacent holes. Therefore, we separately investigate the liquid behavior in the single and multiple holes aiming to address practical issues and obtain accurate characterizations.

To measure the level of internal liquid (refrigerant), as shown in Figs. 2(b) and 2(c), we prepare a transparent acrylic chamber with a distribution plate at the bottom, wherein the inner liquid is supplied from the top. Water and ethanol are used as the liquids. The physical properties are listed in Table I. (REFPROP 9.1) We use seven SUS-316 plates with different hole diameters and numbers of holes. The geometric factors are listed in Table II. The chamber size for single hole is $0.024 \times 0.020\ \mathrm{m}^2$. The plate thickness used is 1 mm and to prevent leakage, 0.5-mm-thick gaskets are used. The same plates are used for both the single and multiple holes, as shown in Fig. 2(a). Additionally, whereas a normal circulation pump is used for multiple holes, a syringe pump is used for a single hole.

Keeping the flow rate constant, we obtain images and videos of the liquid behaviors with surface height change, recorded by a high-speed camera at a frame rate, $500\ \mathrm{s}^{-1}$. The flow rate ranges from 0 to 140 ml/min for a single hole and from 0 to 10 ml/min for multiple holes.

Subsequently, we construct theories to predict the transient height change and the equilibrium height of the liquid surface in the distributor with a single hole. Also, we employ the theories to compare with the experimental results of the multiple holes.

III. SINGLE HOLE DISTRIBUTOR ANALYSIS

To understand the effect of flow rate change on the liquid behavior, we first consider the simplest case of a distributor with a single hole instead of the common multiple holes such as 60 or 80. The liquid is introduced into the vessel at a constant flow rate. The liquid surface

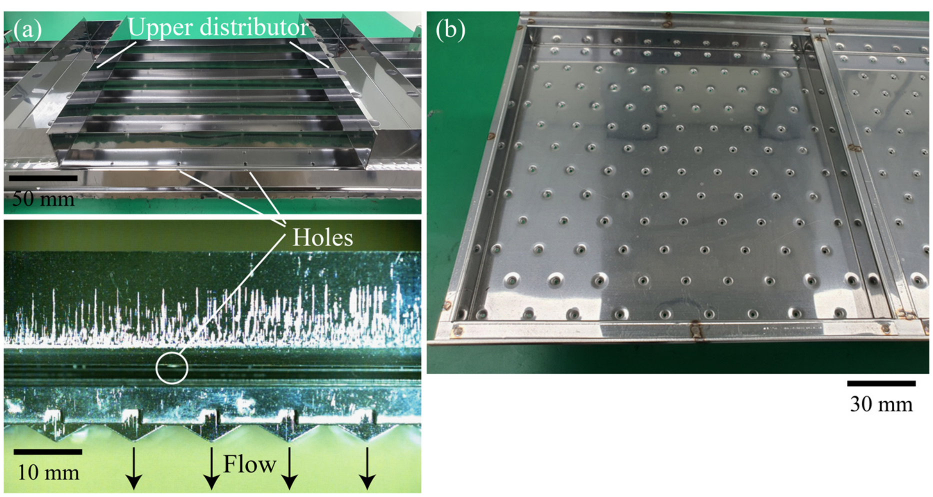

FIG. 1. Common types of distributors for evaporators. (a) A distributor with triangular guides. Liquid is supplied from the upper distributors and moves through the small holes at the bottom of chambers. (b) Another type of distributor with numerous circular holes.

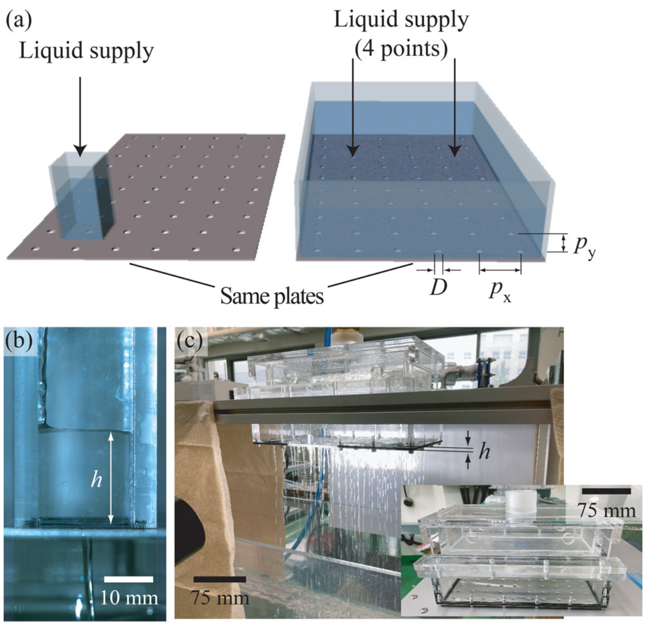

FIG. 2. (a) Schematics of experimental setup for distribution tray with single and multiple holes with liquid supplied from the top. The hole diameters and plates employed are identical in both cases. (b) and (c) show the photographs of the experimental setup for distributors with single and the multiple holes, respectively. The liquid temperature is approximately 25 °C.

TABLE I. Physical properties of the liquid used in experiments (1 atm, 25 °C).

	Density (kg/m^3)	Surface tension (N/m)	Viscosity (Pa s)
Water	997.0	0.072	0.00089
Ethanol	875.1	0.022	0.0011

TABLE II. Geometric conditions of bottom plates used in experiments.

	D (m)	p_x (m)	p_y (m)	N
A	0.0015	0.024	0.020	60
B	0.0017	0.024	0.020	60
C	0.0020	0.024	0.020	60
D	0.0025	0.024	0.020	60
E	0.0030	0.024	0.020	60
F	0.0020	0.040	0.040	15
G	0.0020	0.030	0.030	21

height, h, is measured from the bottom end of the hole, and the experiment began with no liquid, $h = 0$.

The liquid does not flow out through the hole until there is sufficient liquid on the tray. In this case, the hydrostatic force is balanced by the capillary force due to the surface tension at the holes. As the amount of liquid is increased, the radius of curvature of the liquid surface at the hole increased. The deformed surface could withstand hydrostatic pressure until a certain liquid level. When the pressure exceeded the Laplace pressure, the liquid flows out through the hole. Capillary number, the ratio of the viscous shear force and the surface tension, approximately ranges from 10^{-4} to 10^{-2}, in these experiments. Also, the Reynolds number approximately ranges from 100 to 1200. So, the viscous effect is negligible.

The same experiment was performed for two holes. Owing to manufacturing imperfections, the microscopic geometries of the two holes are not identical, resulting in different Laplace pressures depending on the surface configurations. Owing to energetic stability, the liquid pass through one hole only, rather than through both holes simultaneously. Once the flow started, the hole with flow is not changed until the flow is finished. Since the major liquid behaviors of the two holes are almost the same with the single hole, we perform experiments for a single hole only.

At the beginning of the outflow, the surface tension does not play an important role any more, since the liquid surface in a hole that withstands the hydrostatic force is broken. From this moment, the hole is open so that the liquid flows, referred to as *Open* regime. The major pressures that affect the flow are hydrostatic pressure, $\sim \rho g h$ and the inertial force, $\sim \rho U^2/2$. Balancing the two forces, we derive the outlet flow speed, $U \sim \sqrt{gh}$, obtaining similar results from the Toricelli's theory.[7] This is valid only when $A \ll A_0$, with A_0 and A being the cross-sectional areas of the vessel and bottom hole, respectively.[8] We use the similar sign ($\sim$), instead of the equality ($=$) because the exact flow speed profile at the hole is not yet determined. If the flow is fully developed, its speed shows a parabolic profile, where the maximum speed is twice the mean speed. Considering the Bernoulli equation for a streamline from the liquid surface to the hole center, the maximum speed is equal to $\sqrt{2gh}$. However, the flow cannot be fully developed, owing to the plate thickness, substantially smaller than the entrance length. Given the difficulty to know the exact speed profile at the hole, we obtain scaling laws with an empirical coefficient rather than exact solutions. Thus, using a proportional constant c, the outlet flow speed is approximately $\approx c\sqrt{gh}$.

When the outflow begins, if the outlet flow rate is higher than the supply flow rate, the liquid surface level decreases. Figure 3 shows the time sequential images of the surface level change. The speed of the level decreases gradually, because the hydrostatic pressure that drives the flow decreases as the liquid level decreases. The flow initially occurs as a jet, but is split into several consecutive droplets when the flow is reduced. This phenomenon, referred to as Rayleigh–Plateau instability, occurs because of the minimization of the surface energy.[17–19] In other words, when the flow rate is considerably low, the jet width is too small to maintain the cylindrical surface shape because of the surface energy. Therefore, the jet breaks into several droplets. Additionally, when the flow rate is so low that the inertial force is insufficient to break the surface, the flow cannot remove the droplets hanging on the bottom of the plate.

Once the outlet flow stops, it does not start for a certain period, because the hydrostatic pressure is too small to drive the flow and overcome the Laplace pressure of the hole. However, owing to the constant liquid supply, the surface level continues to increase until the flow resumes. As shown in Fig. 3(a), the surface level increased at a constant rate and flow resumes once the hydrostatic pressure became sufficiently large. Thereafter, the surface level appeared to increase and decrease periodically.

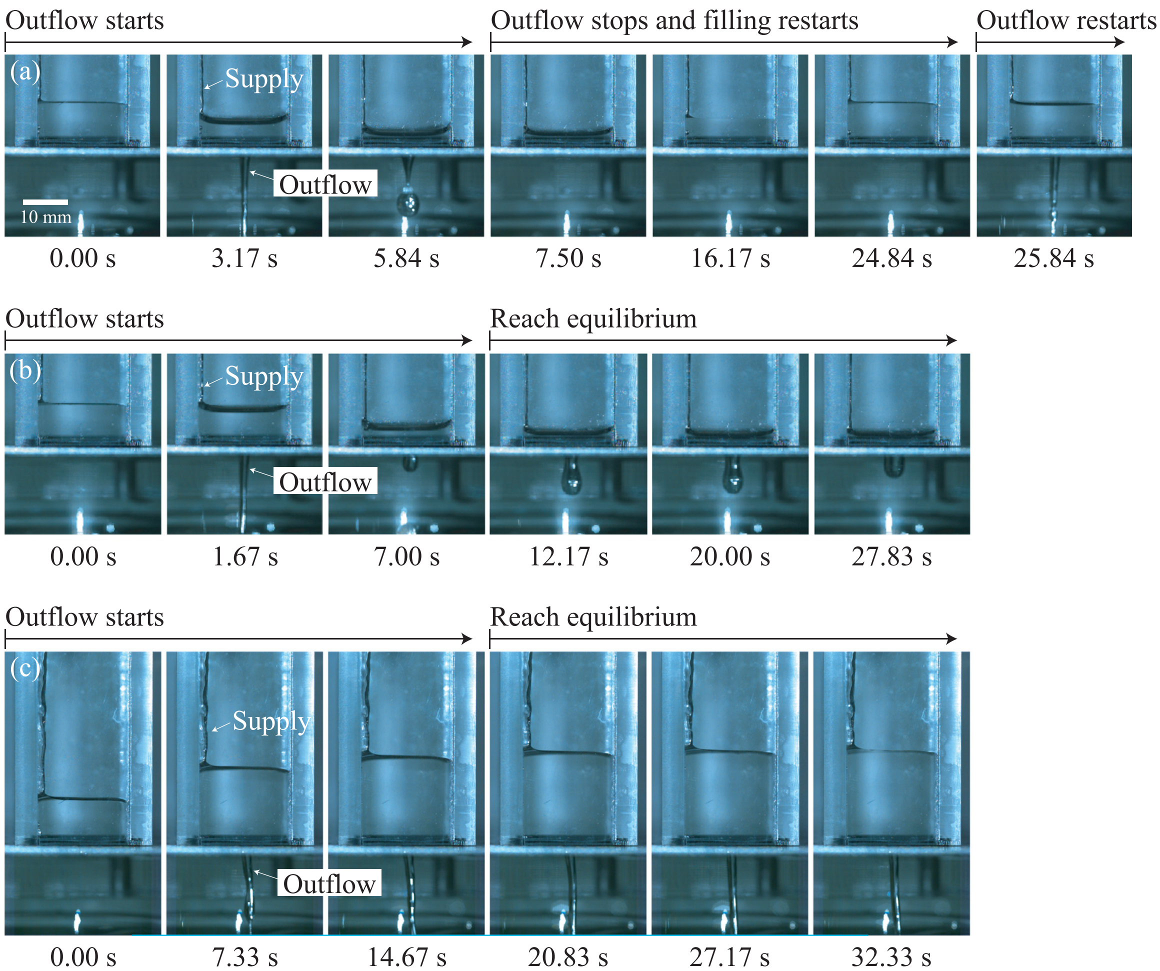

FIG. 3. Time sequential images of the three different cases, (a) when the periodic perturbation occurs ($D = 2.0$ mm, $Q = 10$ ml/min), (b) when $h_{eq} < h_J$, ($D = 2.0$ mm, $Q = 25$ ml/min), and (c) when $h_{eq} > h_J$, ($D = 2.0$ mm, $Q = 100$ ml/min) with h_{eq} and h_J being the equilibrium height and the Jurin's height, respectively.

For a high supply flow rate, the surface level is increased, regardless of the outflow. The surface level increases until the outlet and supply flow rates are balanced. To understand the level change, we focus on the volume conservation in the vessel. Considering the supply (Q) and outflow ($c\sqrt{gh}A$) rates, the volume change rate denoted as $\dot{V} = A_0(\mathrm{d}h/\mathrm{d}t)$ yields

$$\frac{\mathrm{d}h}{\mathrm{d}t}A_0 = Q - c\sqrt{gh}A. \tag{1}$$

Considering $\mathrm{d}h/\mathrm{d}t = 0$, we obtain the equilibrium height

$$h_{eq} = \frac{1}{c^2 g}\left(\frac{Q}{A}\right)^2. \tag{2}$$

As shown in Fig. 4(a), the equilibrium height is monotonically increased as the supply flow rate is increased. The height shows a linear relation with the square of Q/A, [see Fig. 4(b)] for which the slope of the best fitting line is 1.0036, being identical to $1/c^2$. Thus, $c \approx 1.00$. In previous studies, we have found similar results to Eq. (2).[7,8] In a paper by Otto and Mungan, a steady-state solution gives the same results as this paper, except for the coefficient c. In another study, the value of c ranges approximately from 0.71 to 1.13 (for $k = 2$, and $k = 8$ in Ref. 7), which is in agreement with our estimate.[7] Employing characteristic time, $\tau = h_{eq}A_0/Q$, we establish the dimensionless form of Eq. (1) with normalized terms, $H = h/h_{eq}$ and $T = t/\tau$, as follows:

$$\frac{\mathrm{d}H}{\mathrm{d}T} = 1 - H^{1/2}. \tag{3}$$

Figure 5 shows the measured liquid surface level change and curve predicted by Eq. (3). The experimental data and the theoretical results suitably agree for different hole size and supply flow rate.

To estimate the perturbation on the liquid surface, we create waves by dropping liquid droplets directly onto the middle of the surface. We additionally attempt to minimize the perturbation by introducing the liquid supply along the walls. We observe that the surface wave locally affects the stability of the jet by changing the instantaneous hydrostatic pressure, but this effect is negligible.

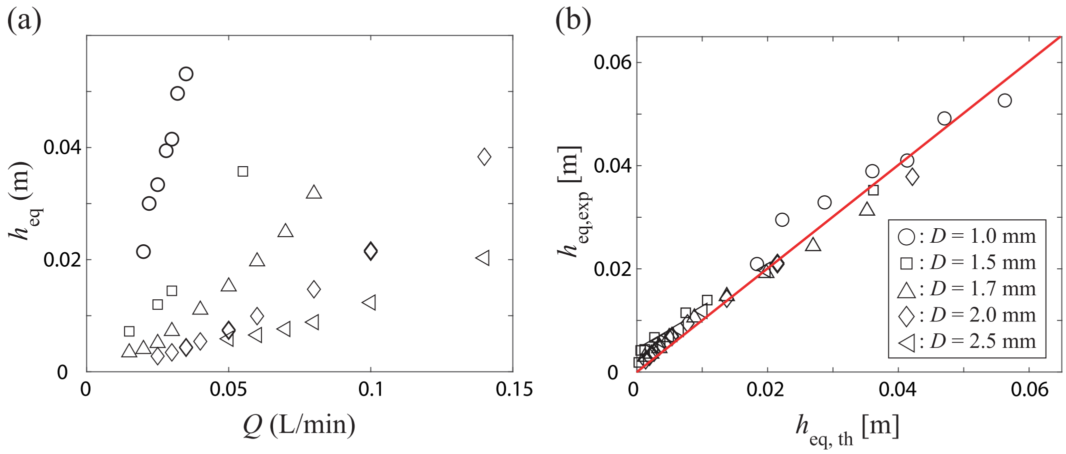

FIG. 4. (a) Equilibrium height of liquid surface according to flow rate, Q. (b) Measured equilibrium height ($h_{eq,\,exp}$) according to Eq. (2) for different hole diameters. The slope of the best fitting line is approximately 1.00.

In industry, the distributor in an evaporator is operated under a sufficiently high flow rate. Thus, we also focus on the high flow rate. As the surface level is zero at the beginning of the experiment, the surface height increases until it reaches Jurin's height for hole diameter, D.[20] Jurin's height is given by $2\cos\theta\sigma/(\rho g r)$, with θ being the contact angle between the liquid and structure. This relation is usually used in the capillary rise problem to predict the spontaneous rise height due to surface tension against gravitational force. In this study, we use the relation for the height that surface tension can withstand at a specific hole diameter. However, as the contact line touches the edge of the structure, the angle at which the force is applied cannot be accurately known. Thus, we assume the angle almost the same in all cases and estimate the Jurin's height, as $\approx 4\sigma/(\rho g D)$, derived by balancing hydrostatic pressure and Laplace pressure on the hole.

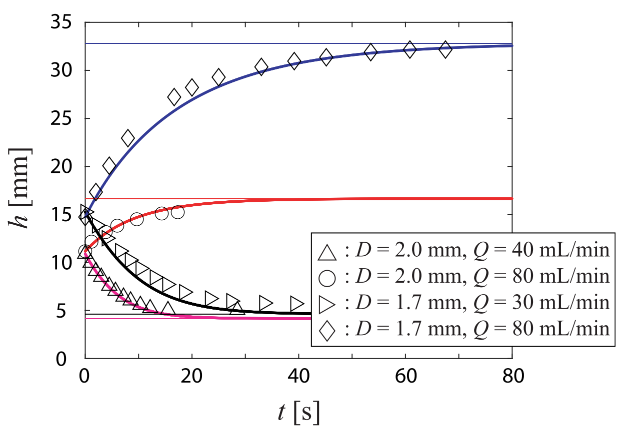

FIG. 5. Comparison between Eq. (3) and the experimental data points. The starting point ($t = 0$ s) indicates the outflow onset when filling is completed. The initial height is approximately equal to Jurin's height, which is determined by the hole diameter. The curves and lines represent the transient height change and the equilibrium height predicted by Eq. (3). The magenta, red, black, and blue curves and lines correspond to ($D = 2.0$ mm, $Q = 40$ ml/min), ($D = 2.0$ mm, $Q = 80$ ml/min), ($D = 1.7$ mm, $Q = 30$ ml/min), and ($D = 1.7$ mm, $Q = 80$ ml/min), respectively.

Figure 5 shows the height change depending on the elapsed time after reaching h_J. If the equilibrium height is larger than h_J, the surface height continues to increase. Otherwise, the surface height is decreased until it reaches the equilibrium height. The two regimes are referred to as *Upper surface* and *Lower surface*, respectively. Considering the transition occurs when $h_{eq} \sim h_J$ and employing Eq. (2), the threshold flow rate, Q_{th}, is scaled as follows:

$$Q_{th} \sim A\sqrt{\sigma/(\rho D)}. \tag{4}$$

Figure 6 shows the relation between Q_{th} and the experimental measurements. The plot includes the two regimes with a separated line, $Q_{th} \sim A\sqrt{\sigma/(\rho D)}$. The line (red) shows good agreement with the experimental data.

Additionally, we analyze the shape of the liquid for $h_J > h_{eq}$, that is, the *Upper surface* regime. When the flow rate is low, droplets are observed, but it is changed to a jet when the flow rate is high. Figure 6 shows the liquid shape change depending on flow rate, where the classification is introduced in other studies.[21,22] The liquid behavior is determined by the inertial force and surface tension. When the flow speed is sufficiently high, the inertial force is large relative to the surface tension, which maintains the shape of the jet, whereas when the speed is low, the small inertial force cannot overwhelm the surface tension, which deforms the jet to the droplets by the aforementioned Rayleigh–Plateau instability. To derive the conditions of behavior change, we balance the relevant forces, as $\rho U^2 \sim \sigma/D$. Using $Q/A = U$, the threshold flow rate is scaled as $Q_{th,sh} \sim A\sqrt{\sigma/(\rho D)}$. The two regimes are plotted in Fig. 6. The transition is delineated by the blue line, $Q_{th,sh}$, which shows good agreement with the experimental data.

As shown in Fig. 5, equilibrium state is slowly reached when the flow rate is high. To quantitatively investigate the time required to reach equilibrium state, we derive an analytic solution of Eq. (3). In the equation, $\mathrm{d}H/(1 - H^{1/2}) = [1/(1 - H) + (-2 + 1/(1 - H^{1/2}) + 1/(1 + H^{1/2}))/(2H^{1/2})]\mathrm{d}H$. By integrating both sides from (T_0, H_0) to (T, H), we obtain the following equation:

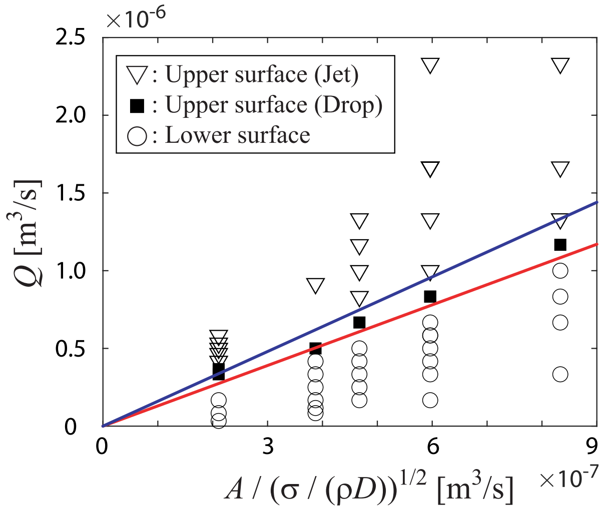

FIG. 6. Three liquid behaviors under different hole diameters and flow rates. The regimes are separated by the scaling law, $\sim A/\sqrt{\sigma/(\rho D)}$. The red and blue lines indicate $Q_{th} \sim A\sqrt{\sigma/(\rho D)}$ with different slopes. The slopes of the best fitting lines are approximately 1.3 (red) and 1.6 (blue), respectively.

$$T - T_0 = -2\left(H^{\frac{1}{2}} - H_0^{\frac{1}{2}}\right) + 2\ln\frac{1 - H_0^{\frac{1}{2}}}{1 - H^{\frac{1}{2}}}. \quad (5)$$

When $T_0 = 0$ and $H_0 = 0$, Eq. (5) becomes $T = -2H^{\frac{1}{2}} - 2\ln\left(1 - H^{\frac{1}{2}}\right)$, corresponding an analytic solution of Eq. (3). Accordingly, as H approaches 1 (h approaches h_{eq}), T tends to be infinity. Theoretically, it implies that it takes infinite time to reach the equilibrium state. In practice, we can derive T for $H = 0.99$. Denoting $T^* = T(H = 0.99)$, $T^* = 8.60$. Considering $t^* = T^*\tau$ and $h_{eq} = Q^2/(gA^2)$ by Eq. (4), $t^* = 8.60QA_0/(gA^2)$. Denoting the number of holes, N, when we expand it to the cases of the multiple holes,

$$t^* = T^*\frac{QA_0}{gN^2A^2} = T^*\left(\frac{h_{eq}}{g}\right)^{1/2}\frac{A_0}{A}\frac{1}{N^2}. \quad (6)$$

Here, we focus on the parametric dependency of the required time to reach equilibrium. Since $A \propto D^2$, $t^* \propto Q/A$, $t^* \propto A_0$, $t^* \propto 1/D^4$, $t^* \propto 1/N^2$, and $t^* \propto h_{eq}^{1/2}$, respectively. Figures 7(a) and 7(b) show the numerical results of the required time with respect to $Q/(NA)$ and D, respectively. As infinite time is theoretically required for complete convergence, we focus on the cases of $H = 0.99$, 0.9, and 0.8, instead of $H = 1$. According to the plots, a longer time is required to reach equilibrium as the supply flow rate increases or the hole diameter decreases. Especially, the hole diameter has a more considerable effect on the required time. Indeed, it is observed in Fig. 5 that the required time for the convergence is much larger when $D = 1.7$ mm, than $D = 2.0$ mm. Thus, in the actual application, if we design a distributor with too small holes to keep the water level high, a long time may be necessary to reach the desired performance. Thus, in the application to the distributor in a falling-film evaporator, a smaller hole diameter causes a longer reaction time when changing the supply flow rate. In contrast, a larger the hole diameter reduces the time required to reach equilibrium. Furthermore, as the surface tension decreases and the density increases, the time is decreased.

IV. MULTIPLE HOLE DISTRIBUTOR ANALYSIS

To investigate the liquid behavior in a realistic distributor, we consider 60 holes. The pitch and the diameter of the holes are same as those of evaporator distributors used in industry. We aim to analyze the actual phenomena by visualizing the flow and measuring the height. The liquid is supplied to the empty vessel and the surface height is measured.

We first observe the outflow through the holes, as shown in Fig. 8. As liquid is provided into the distributor, the liquid surface in a hole becomes bulged due to the hydrostatic pressure. When the pressure is sufficiently large, the bulged surface becomes a droplet, falling toward the tubes. When the flow rate is low, droplets are formed at some holes, and flow from a few of them. As the flow rate increases, the number of droplets in holes also increases. This is referred to as *Partial falling* regime. See Figs. 8(a)–8(c) and the inset of Fig. 9. In the *Partial falling* regime, since there exist a few holes where flow does not occur, the hydrostatic pressure is locally balanced with the Laplace

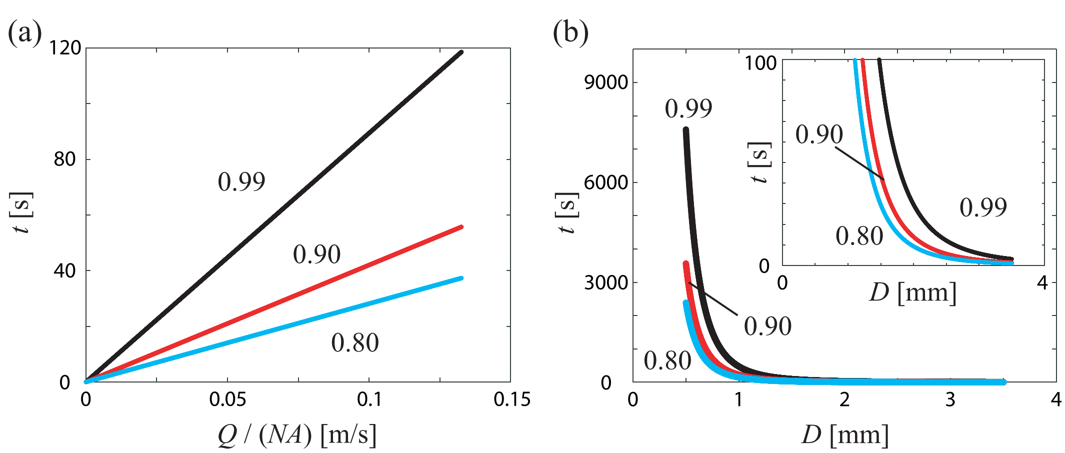

FIG. 7. Required time (t) of (a) $Q/(NA)$ and (b) D, derived by Eq. (6), to reach $H = 0.8$, 0.9, and 0.99, respectively. Other conditions: (a) $D = 2.0$ mm, $N = 1$ and (b) $Q = 0.100$ l/min, $N = 1$.

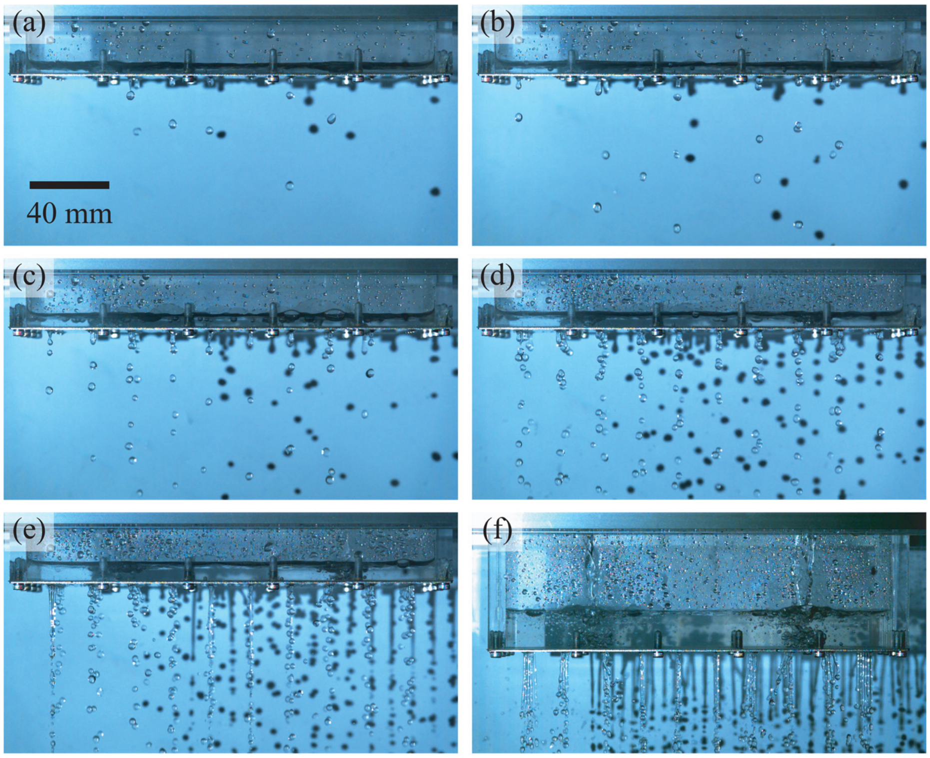

FIG. 8. Visualization of the liquid behavior for $D = 2.0$ mm and $N = 60$, depending on the flow rates, (a) 0.2l/min, (b) 0.4 l/min, (c) 0.8 l/min, (d) 2.2 l/min, (e) 3.4 l/min, and (f) 5.0 l/min.

pressure made at the holes, $\approx 4\sigma/D$. The height approximately corresponds to Jurin's height, $\approx 4\sigma/(\rho g D)$. Thus, the height does not notably increase if those holes are existed, although flow rate is increased, which causes a plateau in the plot, as shown in the inset of Fig. 9.

In the plateau, the height is slightly increased, due to minor hole shape deviations from the manufacturing process, which cannot be perfectly removed. For instance, a small protrusion toward the inside

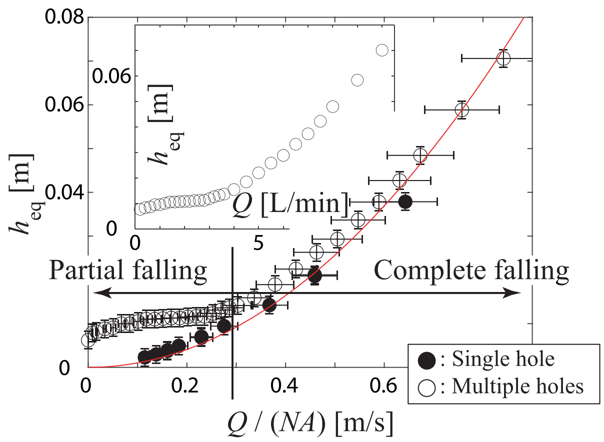

FIG. 9. Equilibrium height according to flow speed, $Q/(NA)$, for the single hole and the multiple holes ($N = 60$). The hole diameters are the same, $D = 2.0$ mm. The inset shows the surface level according to the flow rate for the same multiple holes. ($D = 2.0$ mm and $N = 60$).

of the hole interferes with the droplet formation when bulging begins, because the solid surface is not highly hydrophilic. Thus, even when droplets exit easily from some holes, the liquid does not exit from other holes. When more liquid is provided such that the height increases, droplets start to flow out through more holes. In other words, a larger hydrostatic pressure is necessary to cause flow in some holes. Hence, the equilibrium height is slightly increased as more holes with droplets appear.

As the liquid is supplied further, the liquid flows from all of the holes. From this point, the unit flow rate from each hole is considerably increased. Simultaneously, the surface level starts to be increased remarkably, as shown in Figs. 8(d)–8(f). This is referred to as *Complete falling* regime. In this regime, the falling frequency becomes much higher, and water droplets are connected to form a jet, as shown in Fig. 8(f).

Returning to the case of a single hole, when the liquid is initially supplied, it does not exit through the hole until it reaches Jurin's height. However, when more liquid is provided such that the hydrostatic pressure exceeds the surface tension, the liquid flows out through the hole. Once the liquid flows through the hole, the surface tension ceases to block the flow. Thus, regardless of a low flow rate, the flow occurs relatively quickly to decrease the surface level, as shown in Fig. 3. Under a perturbation, when the liquid reaches the equilibrium state, the surface level converges to the equilibrium height.

On the other hand, different characteristics are observed at a low flow rate for multiple holes. When liquid is initially supplied, it does not exit through the holes until the surface reaches Jurin's height, being consistent with the single-hole case. However, when more liquid is supplied, perturbation does not occur. Once the surface level reaches

Jurin's height, it remains constant, even though more liquid is supplied. The liquid flow through multiple holes and the unit flow weakens at a low flow rate, in the partial falling regime, as shown in Fig. 9. Typically, if the hole diameter is not very large, the unit flow forms consecutive droplets, but not a jet, owing to the surface tension.

When the outflow rate is increased to establish the complete falling regime, the liquid behavior is approximately the same as that for a single hole. By balancing the two dominant forces, the hydrostatic force and the inertial force, we derive $(Q/A)^2 = c^2 g h_{eq}$, which is equivalent to Eq. (2). Considering a streamline from the liquid surface to the hole, the relation is also derived by the Bernoulli equation, being similar to Toricelli's theory. The experimental measurements confirm that constant c is approximately unity, as derived above.

The red line in Fig. 9 indicates the theoretical curve derived by Eq. (2). Using all the experimental data, we compare the raw data and the theoretical results, as shown in Figs. 10(a) and 10(c). The experimental raw data are observed to be scattered, whereas the entire data points of high flow rate in non-dimensional form are coalesced onto one single line. The slope is 1.0035, similar to the slope in Fig. 3. The experimental conditions are listed in Fig. 10(d).

To sustain a jet through one hole for a long time, the flow must be stable without external interference. Notably, in a falling-film evaporator, the jet flow velocity according to the height is uniquely determined. For multiple holes, if the flow in an adjacent hole suddenly occurs and the supply is reduced, the flow rate immediately decreases and the jet shrinks. The mitigation of the jet indicates that a reduced flow rate and relatively strong local surface tension, which gives rise to a pendant (fiber) form of flow.[21] In this case, the flow occurs in droplet and the forces applied to the system achieve a new equilibrium. Hence, the inertial force becomes relatively small owing to the sudden velocity drop, and the dominant equilibrium forces are the hydrostatic force and surface tension. Therefore, the height is near Jurin's height.

For a single hole, there is one supply and one outlet. For multiple holes, the number of outlets outnumbers the supply. Despite the same unit velocity, $Q/(AN)$ in these two cases, multiple holes induce a wide range of internal flows in the distributor chamber. As these flows locally increase or reduce the pressure above the holes, some droplets begin to be formed before the water surface reaches Jurin's height, h_J. Indeed, droplets already spread relatively widely at the lower part of several holes before the height of the water surface becomes sufficiently high. As the diameter of the growing droplet at the bottom is larger than the hole diameter, the capillary force, which blocks the liquid outflow through the holes, is typically smaller than that when the droplet

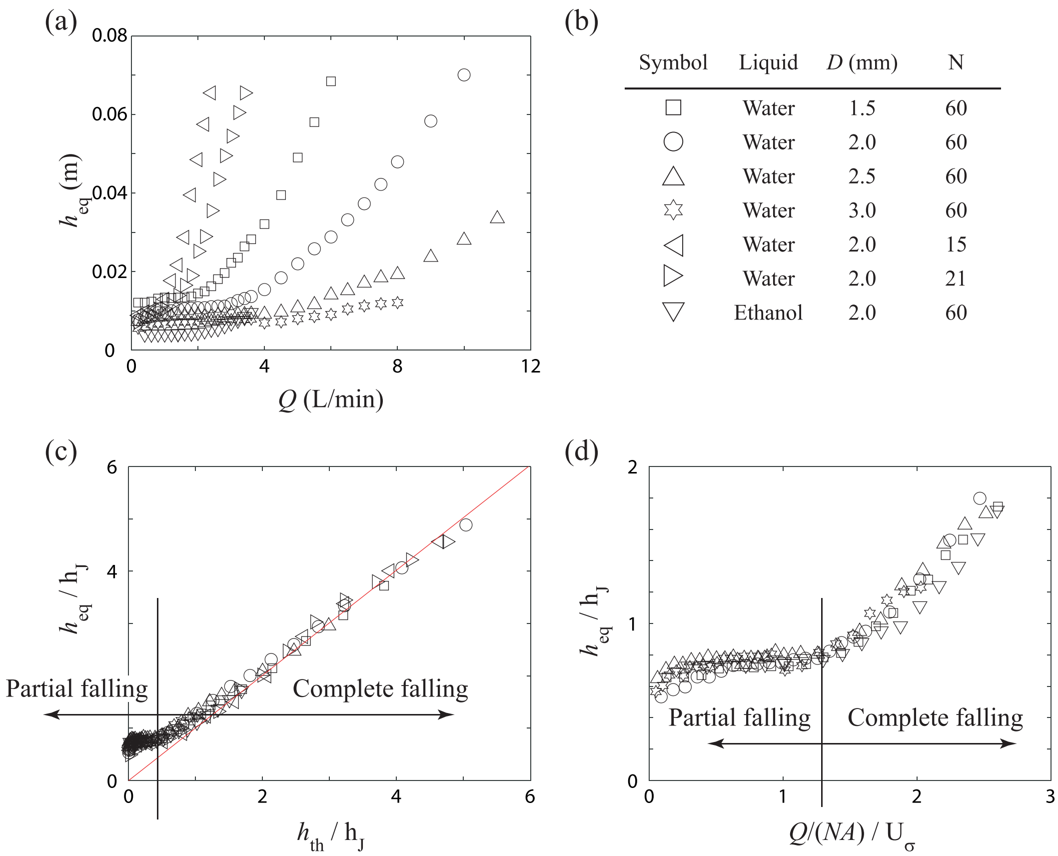

FIG. 10. (a) Equilibrium height of liquid surface according to flow rate, Q. (b) Experimental conditions for the symbols. (c) The normalized measured equilibrium height (h/h_J) plotted according to Eq. (2). The slope of the best fitting straight line is approximately 1.00. The transition between partial filling and complete filling arises at $h_{th}/h_J \approx 0.423$. (d) The normalized height vs the normalized unit flow speed, $Q/(NA)/U_\sigma$. The transition between the two regimes arises at $Q/(NA)/U_\sigma \approx 1.3$.

contact line is located at the vicinity of the hole boundary. The droplet continues to grow until it becomes excessively heavy to adhere to the plate bottom, resulting in droplet falling. Consequently, multiple holes do not show strong jets, and the water level does not decrease, differing from the single hole case.

Various equilibrium points may exist at a flow rate at which the theoretical equilibrium height [see Eq. (2)] is lower than Jurin's height. This is because the contact line at the bottom is not determined by a single hole diameter, which is directly associated with the capillary force against the hydrostatic force of the water level. Hence, the water level or droplet behavior is not determined by one flow rate. Additional studies are needed for more accurate predictions, such as the quantitative analysis of the flow inside the distributor while changing the number and size of holes, as well as the distance between holes.

Considering that the transition between the two regimes occurs when the equilibrium height exceeds Jurin's height, we derive a threshold for certain h_{eq}/h_J. Using the results for a single hole shown in Fig. 6, we obtain the threshold as $Q_{thre}/(NA)\sqrt{\sigma/(\rho D)} \approx 1.3$, which corresponds to the slope of the red line in the plot. Employing $h_J = 4\sigma/D$ and $h_{th,eq} = (Q/A)^2/g$, $h_{th,thre}/h_J = 0.423$. As the line indicates the regime transition between *lower surface* and *upper surface* for a single hole, the threshold corresponds to the unit flow inertial force beginning to overcome the resisting surface force. Considering that the speed generated by the surface tension is scaled as $U_\sigma \sim \sqrt{\sigma/(\rho D)}$, we have $Q/(NA)/U_\sigma \sim \sqrt{h_{th}/h_J}$. The threshold value is approximately 1.3. The threshold also indicates when the equilibrium height is above Jurin's height for multiple holes. Therefore, the threshold indicates the end of the plateau in which the surface height is close to Jurin's height. In addition, as the number of flowing holes is maximally increased, it is the end of the partial falling regime and the beginning point of the complete falling, as shown in Figs. 10(c) and 10(d), where a clear separation is observed. The data points in Fig. 10(c) start to coalesce onto the line of Eq. (2). The definition of the Jurin's height is $h_J = 2\cos\theta\sigma/(\rho g D/2)$, as we mentioned. We note that the contact line lies on the edge at each hole. Since the contact angle at the edge is not uniquely determined, we use the Jurin's height with $\theta = 0$. Thus, the h_J is overestimated, which makes the saturation at the plateau smaller than 1. Considering the threshold in Fig. 10(d) is determined by balancing the flow speeds driven by inertial force and surface tension, it can be written as a square root of Weber number, $We = \rho(Q/(NA))^2 D/\sigma$, since $We^{1/2} = Q/(NA)/\sqrt{\sigma/(\rho D)} \sim Q/(NA)/U_\sigma$.

As shown in Fig. 10(c), once liquid flows through all the holes, the data points rapidly converge to the line of the theoretical formulation. In the complete falling regime, the liquid falls as consecutive droplets at a low flow rate. As the flow rate is increased further, the droplets turn into a jet. However, regardless of the form, once the complete falling regime starts, it follows Eq. (2), being consistent with the results for a single hole, as shown in Fig. 6.

In the *partial falling* regime, we observe a plateau, as shown in Figs. 10(c) and 10(d), which means that the hydrostatic pressure is almost the same for different flow rates. Therefore, when designing a distributor for such low flow rate conditions, it should be noted that the liquid supply is not evenly distributed but concentrated through several holes. In this regime, as the flow rate is changed, the number of flowing holes also changes.

V. DISCUSSION

In a falling-film evaporator, surface height control in the distributor considering the device inclination is important.[15] In general, the length of the evaporator in a turbo chiller is approximately 3 m, and that of the distributor is almost the same. Therefore, even if the floor is slightly tilted from the horizontal, the height of the water surface at opposite sides can change greatly. For example, if a 3 m evaporator is tilted approximately 3/1000 rad (0.17°), the height variation of the water surface at opposite sides is approximately 9 mm. Employing the aforementioned Eq. (2), when $D = 2$ mm, if the height of the water surface is approximately 43 mm, the difference in the supply flow rate at both sides is within 10%. If the water level inside the lower distributor becomes very large, an appropriate height should be selected because the distributor may become relatively large compared to the overall size of the evaporator. In other words, if the height of the water becomes very low by increasing the number or diameter of holes, the distributor operation becomes sensitive to inclination.

To design a distributor, the aforementioned required time to reach equilibrium should be also considered. According to the results in Fig. 7, it might take longer time than predicted if the hole diameter or number of holes is very small. Therefore, design parameters of the distributor, such as hole size, pitch, and number of columns, must be carefully selected.

There are two types of falling-film evaporators: those with a feed pump and those without. When a pump is separately installed, despite a small liquid amount being supplied from the condensing unit, increasing the circulating flow rate allows to increase the amount supplied to the heat transfer tubes. Therefore, there are fewer constraints on the selection of the design parameters. However, abundant refrigerant may be required to operate the pump in the evaporation shell, likely increasing the total operation cost. On the other hand, if a pump is not used, design parameters such as the number of heat pipe columns must be carefully selected, because the amount of liquid supplied from the condensing unit is identical to the amount supplied to the heat transfer tubes. If there are an excessive number of columns, the height of the liquid surface in the distributor becomes extremely low, consequently increasing the sensitivity to inclination.

For an evaporator without a pump, the flow rate is often smaller than that with a pump. If the tube length or number of columns is large, the liquid (refrigerant) surface on the distributor can become very low owing to the large bottom area of the distributor. In this case, if the flow rate is very low, the liquid flow may be classified in the *partial falling* regime. If it is in the regime, outflow does not occur at all the holes, and the position of the flowing holes is difficult to be predicted. Consequently, the evaporator performance may be much lower than expected. Unlike the high flow rate region, at the vicinity of the regime boundary, the surface tension becomes important. Therefore, employing the plot in Fig. 10(d), the unit flow speed must be evaluated by comparing with U_σ.

In an actual system, the distributor is installed with several layers of upper distributors, since the working liquid is supplied to a large area from a single pipe connected with a condenser or expansion valve. The upper distributor has a few large holes with a constant gap for uniform supply. Gravity drives the flow, which makes streams from the upper distributors to the final one that has plenty of small holes. When the liquid reaches the final distributor, the liquid pressure of the stream due to the elevation difference reduces the liquid surface level

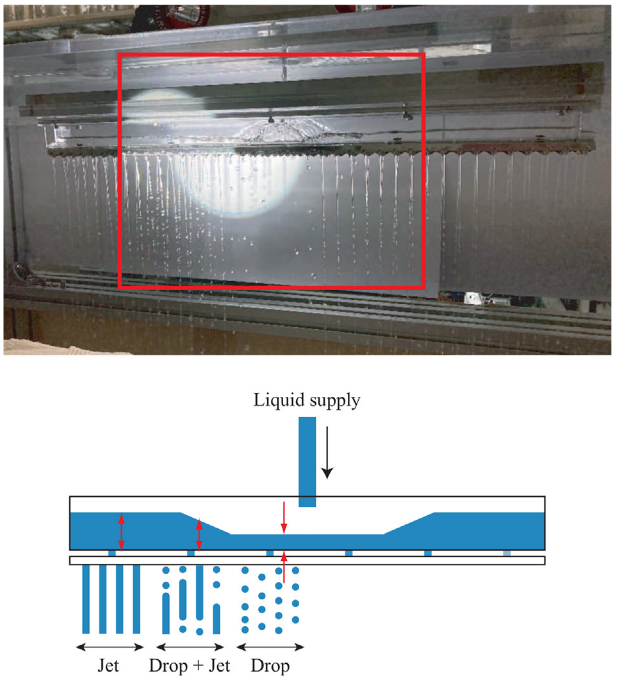

FIG. 11. At the spot where the main stream falls from the upper chamber, the local pressure pushes liquid toward both sides. Owing to the reduced surface height, a local drop in hydrostatic pressure reduces the local supply to the tube part under the distributor. The working liquid in this experiment is ethanol.

locally. At the spot of the liquid stream, the lower surface level creates a locally small hydrostatic pressure at nearby holes. Consequently, the local supply decreases to heat transfer tubes under the final distributor.

To visualize the liquid behavior, we prepare an additional experimental setup with a real distributor of an absorption chiller, as shown in Fig. 11. For the visualization, we manufactured a plexiglass chamber instead of the SUS-316 chamber over the distributor. To emphasize the surface level effect, we supply liquid at a single spot. The total number of holes is 80. Considering the wetting characteristics of the synthetic refrigerants, ethanol is used for the working liquid. As shown in Fig. 11, less liquid is supplied to the chamber where the liquid stream reaches the surface. Not only droplets are observed more than jets, but *partial falling* also occurs, while jets are only observed far from the spot.

The local outflow decrease by the hydraulic effect occurs commonly in the distributor inside the evaporator. The decrease extent is determined by the flow rate, stream size, equilibrium height of the liquid surface in the chamber, chamber width, and liquid properties. To reveal parametric dependencies more accurately, additional quantitative studies are necessary. Considering the frequency of this phenomena in a falling-film evaporator, additional practical evaluations are required, based on various experiments for actual applications.

VI. CONCLUSIONS

In this study, we have focused on the liquid behaviors in the distributor of the falling-film evaporator. To understand the mechanism more accurately, we have performed two different experiments, with single and multiple holes. For a single hole, we have experimentally demonstrated that the equilibrium height is directly proportional to the flow rate. In addition, we have derived the relation between the transient liquid level and final height in equilibrium. Furthermore, by solving the differential equation for level change, we have investigated factors that affect the time taken to reach equilibrium.

For multiple holes, unlike the single hole case, we have observed that the liquid level is not changed even when the flow rate is increased at low flow rates. In contrast, at high flow rates, the liquid level is directly proportional to the flow rate, which is in good agreement with the theory derived for single hole. The fundamental difference between the low and high flow rates is related to whether flow occurs in the entire hole or not. Associated with the single hole cases, conditions for the threshold have been elucidated. Furthermore, we introduced the effect of the supply stream that decreases the local hydrostatic pressure in the chamber, thereby locally reducing the outflow rate through holes.

In a real evaporator, the distributor not only supplies liquid to the heat transfer tubes, but also serves as a passage for supplying gas to a subsequent component (e.g., compressor). Therefore, to avoid moisture invasion in such components, the distributor must be carefully designed. We analyzed problems related to the inclination and hydraulic effect owing to the supply stream at the same time. We expect that the derivations and findings from this study serve as a suitable and practical guideline for improved distributor design.

ACKNOWLEDGMENTS

This work was supported by Korea Evaluation Institute of Industrial Technology (KEIT) and Korea Institute of Energy Technology Evaluation and Planning (KETEP), funded by the Ministry of Trade, Industry and Energy (MOTIE) of the Republic of Korea (Grant Nos. 20010090 (KEIT) and 20208901010010 (KETEP)).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jungchul Kim: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Funding acquisition (lead); Investigation (lead); Methodology (lead); Project administration (lead); Resources (lead); Supervision (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Chan Geun Kim:** Software (supporting). **Hak Soo Kim:** Validation (supporting). **Wookyoung Kim:** Validation (supporting). **Dong Ho Kim:** Funding acquisition (lead).

DATA AVAILABILITY

The data that support the findings of this study are available within the article.

NOMENCLATURE

A	Cross-sectional area of the bottom holes
A_0	Cross-sectional area of the vessel
D	Pore diameter
h	Surface level in a distributor vessel

h_{eq}	Equilibrium height
$h_{eq,\exp}$	Measured equilibrium height
$h_{eq,th}$	Predicted equilibrium height by Eq. (2)
h_J	Jurin's height
H	Normalized height, h/h_{eq}
H_0	Initial H
N	Number of pores
p_x	Longitudinal pitch (center to center)
p_y	Transverse pitch (center to center)
Q	Supply flow rate
Q_{th}	Threshold flow rate between *Upper surface* and *Lower surface* regimes
$Q_{th,sh}$	Threshold flow rate between *Upper surface* (Jet) and *Upper surface* (Drop) regimes
T	Normalized time, t/τ
T_0	Initial T
U	Outlet flow speed through a pore
U_σ	Flow speed driven by surface tension
$\dot{V}$	Volume change rate of the refrigerant in the vessel
μ	Refrigerant viscosity
ρ	Refrigerant density
σ	Refrigerant surface tension
τ	Characteristic time

REFERENCES

[1] W. Goetzler, T. Sutherland, M. Rassi, and J. Burgos, *Research & Development Road Map for Next-Generation Low Global Warming Potential Refrigerants* (U.S. Department of Energy, 2014).
[2] X. Sun and T. Ferris, "The Kigali Amendment's and China's critical roles in evolving the Montreal Protocol," Nat. Resour. Environ. **33**(2), 30–34 (2018).
[3] S. J. Hong, E. S. Wang, and C. W. Park, "Heat transfer characteristics of falling film and pool boiling evaporation in hybrid evaporator in vapor compression system," Appl. Therm. Eng. **153**, 426–432 (2019).
[4] P.-H. Jin, C.-Y. Zhao, W.-T. Ji, and W.-Q. Tao, "Experimental investigation of R410A and R32 falling film evaporation on horizontal enhanced tubes," Appl. Therm. Eng. **137**, 739–748 (2018).
[5] L.-H. Chien and Y.-L. Tsai, "An experimental study of pool boiling and falling film vaporization on horizontal tubes in R-245fa," Appl. Therm. Eng. **31**, 4044–4054 (2011).
[6] G. Ribatski and A. M. Jacobi, "Falling-film evaporation on horizontal tubes—A critical review," Int. J. Refrig. **28**, 635–653 (2005).
[7] S. D'Alessio, "Torricelli's law revisited," Eur. J. Phys. **42**, 065808 (2021).
[8] J. Otto and C. E. Mungan, "Filling and emptying a tank of liquid," Eur. J. Phys. **43**, 055003 (2022).
[9] C. Clanet, "Clepsydrae, from Galilei to Torricelli," Phys. Fluids **12**(11), 2743 (2000).
[10] J. Ferrand, L. Favreau, S. Joubaud, and E. Freyssingeas, "Wetting effect on Torricelli's law," Phys. Rev. Lett. **117**, 248002 (2016).
[11] S. M. Kim, J. I. Kim, S. M. Woo, M. S. Kim, and C. W. Park, "A study on evaporation heat transfer characteristics of heating tubes for falling film type evaporator of absorption chiller or adsorption chiller," Korean J. Air-Cond. Refrig. Eng. **30**(11), 508–517 (2018).
[12] D.-H. Park, H.-S. Ham, and N.-H. Kim, "An experimental study on thin film evaporation of water on enhanced tubes used for evaporators of absorption chillers," Korean J. Air-Cond. Refrig. Eng. **31**(5), 183–192 (2019).
[13] Y. Li, K. Wang, W. Wu, X. Xia, B. Niu, and Z. Zhang, "Study on the effect of refrigerant distributing nonuniformity on the performance of falling-film evaporator with liquid recirculation system," Int. J. Refrig. **82**, 199–211 (2017).
[14] V. Sernas, "Heat transfer correlation for subcooled water films on horizontal tubes," J. Heat Transfer **101**, 176–178 (1979).
[15] M. Numata and K. Kasai, "Heat exchanger," U.S. patent US9513039 (6 December 2016).
[16] K. R. Morison, "Reduction of fouling in falling-film evaporators by design," Food Bioprod. Process. **93**, 211–216 (2015).
[17] D. T. Papageorgiou, "On the breakup of viscous liquid threads," Phys. Fluids **7**, 1529 (1995).
[18] J. Eggers, "Nonlinear dynamics and breakup of free-surface flows," Rev. Mod. Phys. **69**(3), 865–929 (1997).
[19] X. Zhang, "Dynamics of growth and breakup of viscous pendant drops into air," J. Colloid Interface Sci. **212**, 107–122 (1999).
[20] J. Jurin, "II. An account of some experiments shown before the Royal Society; With an enquiry into the cause of the ascent and suspension of water in capillary tubes," Philos. Trans. R. Soc. London **30**, 739–747 (1718).
[21] C. Clanet and J. C. Lasheras, "Transition from dripping to jetting," J. Fluid Mech. **383**, 307–326 (1999).
[22] B. Ambravaneswaran, H. J. Subramani, S. D. Phillips, and O. A. Basaran, "Dripping-jetting transition in a dripping faucet," Phys. Rev. Lett. **93**(3), 034501 (2004).